

BIT 4107-Mobile Application Development

ENHANCED PROFESSIONAL STUDENT LOGBOOK

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Admission Number:BIT/2024/49929

School/Faculty: COMPUTING AND INFORMATICS

Programme:Blood_Donation_App

Academic Year: THREE

Semester:TWO

Lecturer: MR.MICHAEL NYORO

Phone Number:0725104634

WEEK 1: INTRODUCTION TO MOBILE APPLICATION DEVELOPMENT

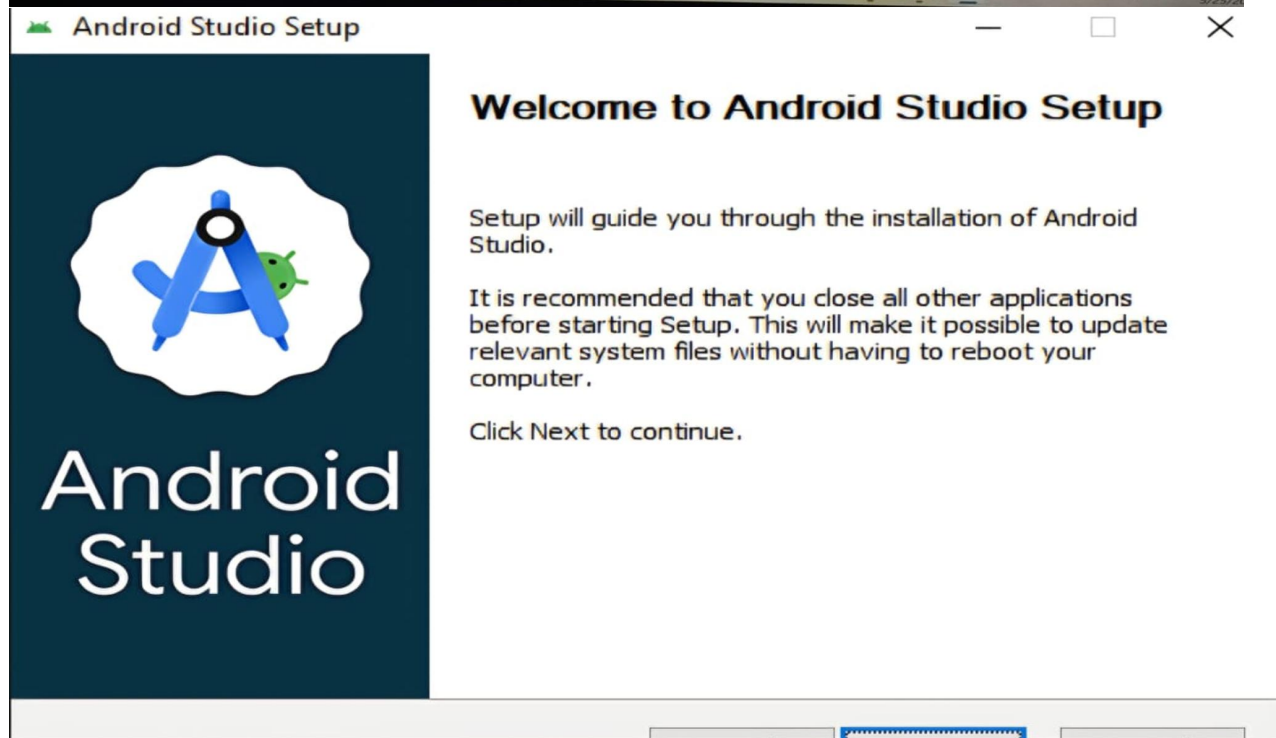
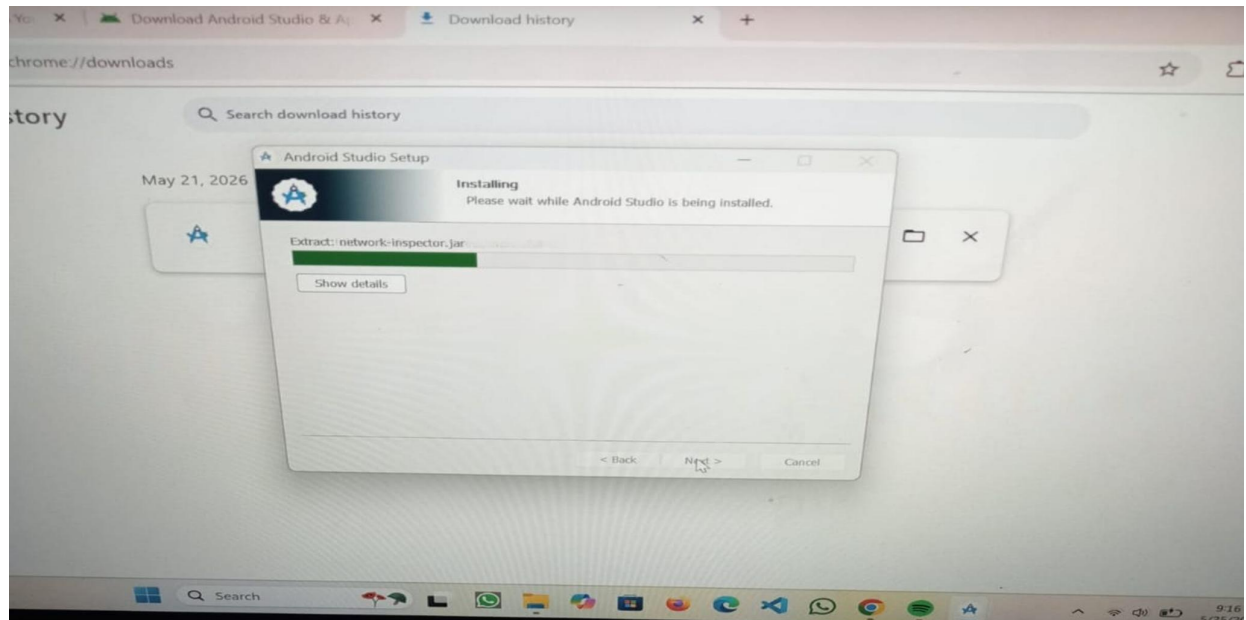
The screenshot displays the Visual Studio Code interface. The top menu bar includes File, Edit, Selection, View, Go, Run, Terminal, and Help. The left sidebar shows the Explorer view with the 'Flutter' extension highlighted. The main editor area displays the 'Flutter' extension details, including the Flutter logo, version v3.42.0, and the Dart Code extension. The bottom panel shows the 'Problems' view with the 'flutter doctor' command output.

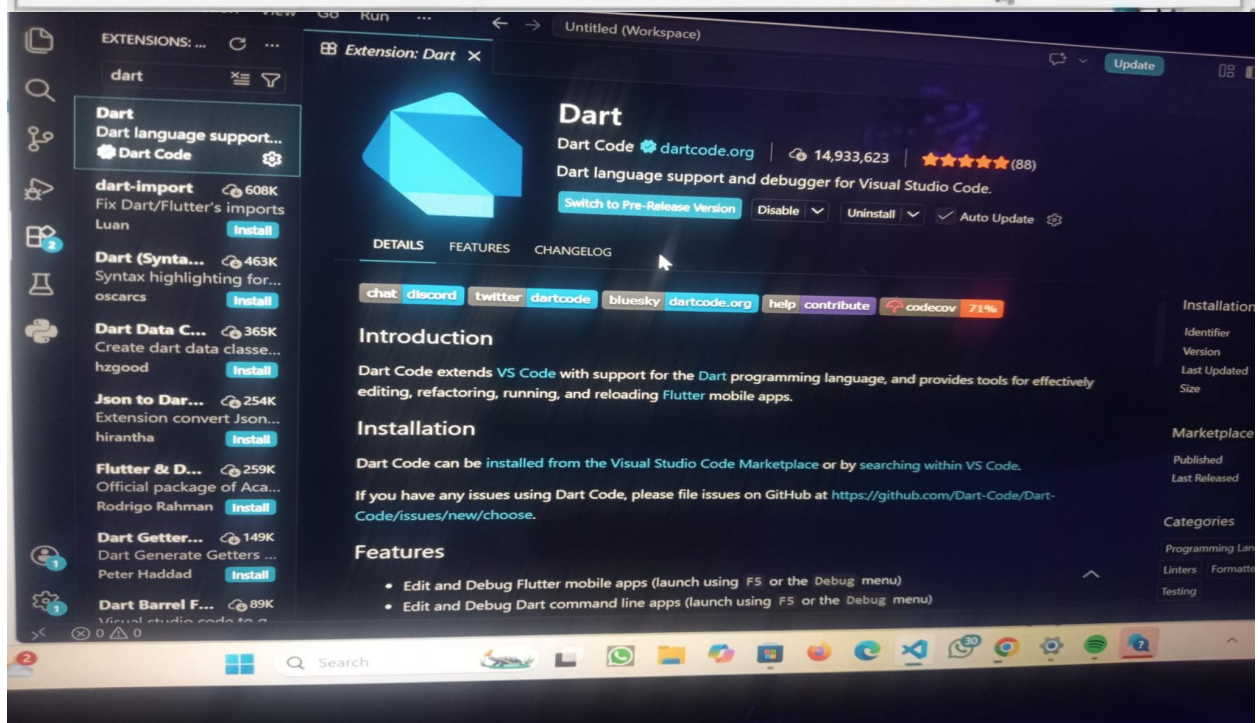
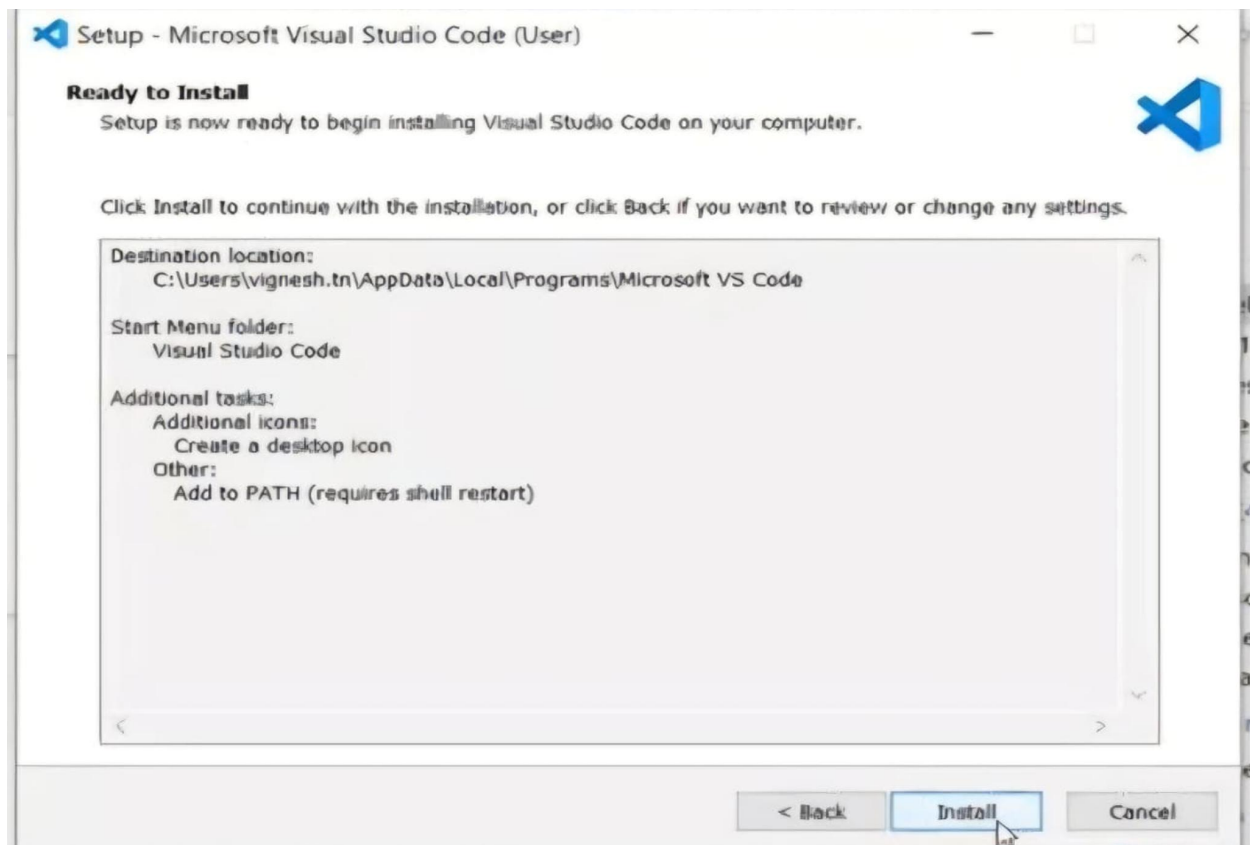
Flutter Extension Details:

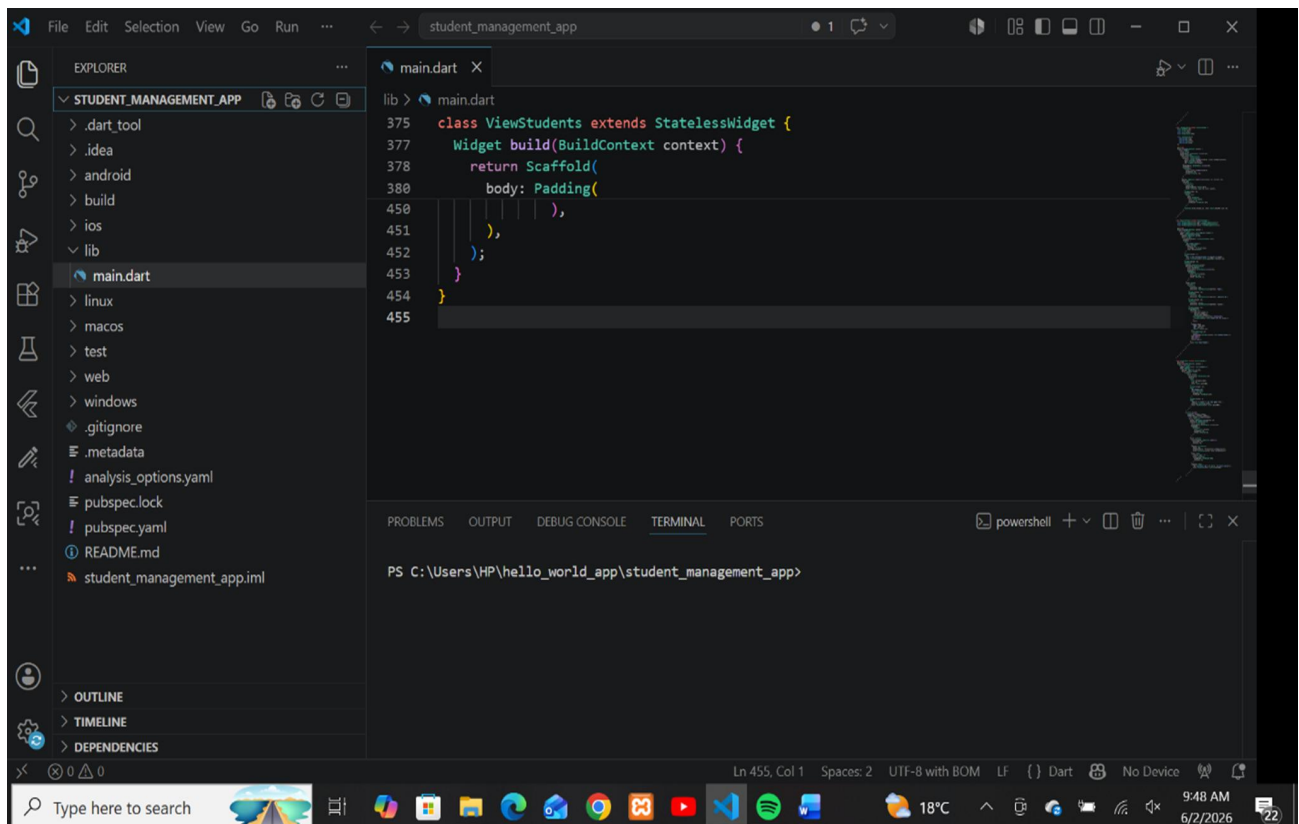
- Flutter v3.42.0
- Dart Code | 4,380,818 | ★
- Flutter support and debugger for Vis...
- Disable | Uninstall | ⚙️
- This extension is enabled globally.

flutter doctor Output:

```
PS C:\Users\HP> flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.44.0, on Microsoft Windows [Version 10.0.19045.6456], locale en-US)
[✓] Windows Version (10 Pro 64-bit, 22H2, 2009)
[!] Android toolchain - develop for Android devices (Android SDK version 36.1.0)
    X cmdline-tools component is missing.
      Try installing or updating Android Studio.
      Alternatively, download the tools from https://developer.android.com/studio#command-line-tools-only and make sure to set the ANDROID_HOME
      environment variable.
      See https://developer.android.com/studio/command-line for more details.
    X Android license status unknown.
```







TOOLS AND SOFTWARE USED

1.Android Studio

2.VS Code

3.Flutter SDK

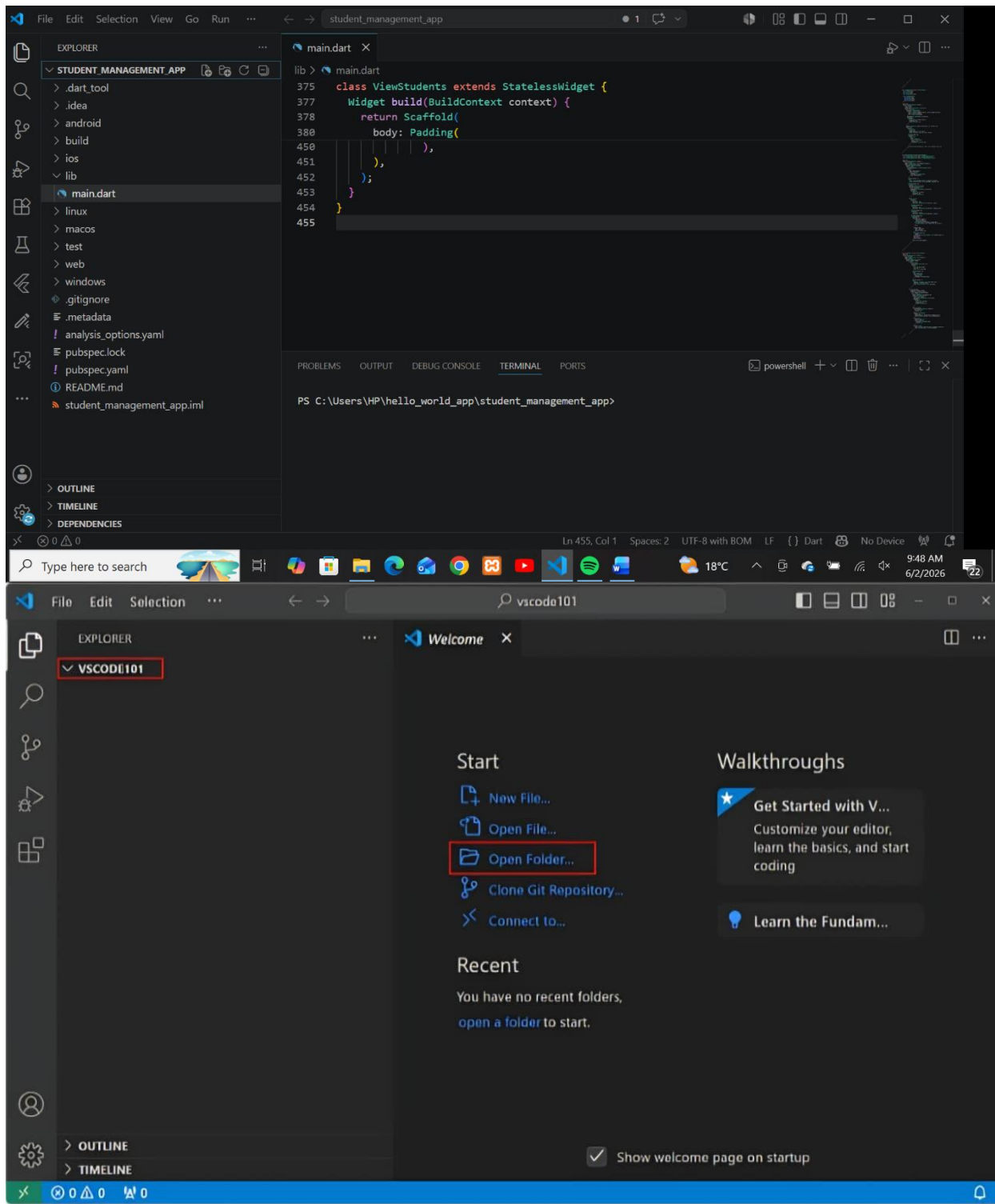
4.Dart

5.Android Phone

PERSONAL REFLECTION

Installing the tools was challenging at first due to errors and configuration issues, but it improved my problem-solving skills. I gained practical experience in setting up development environments and became more confident in using modern software tools for app development.

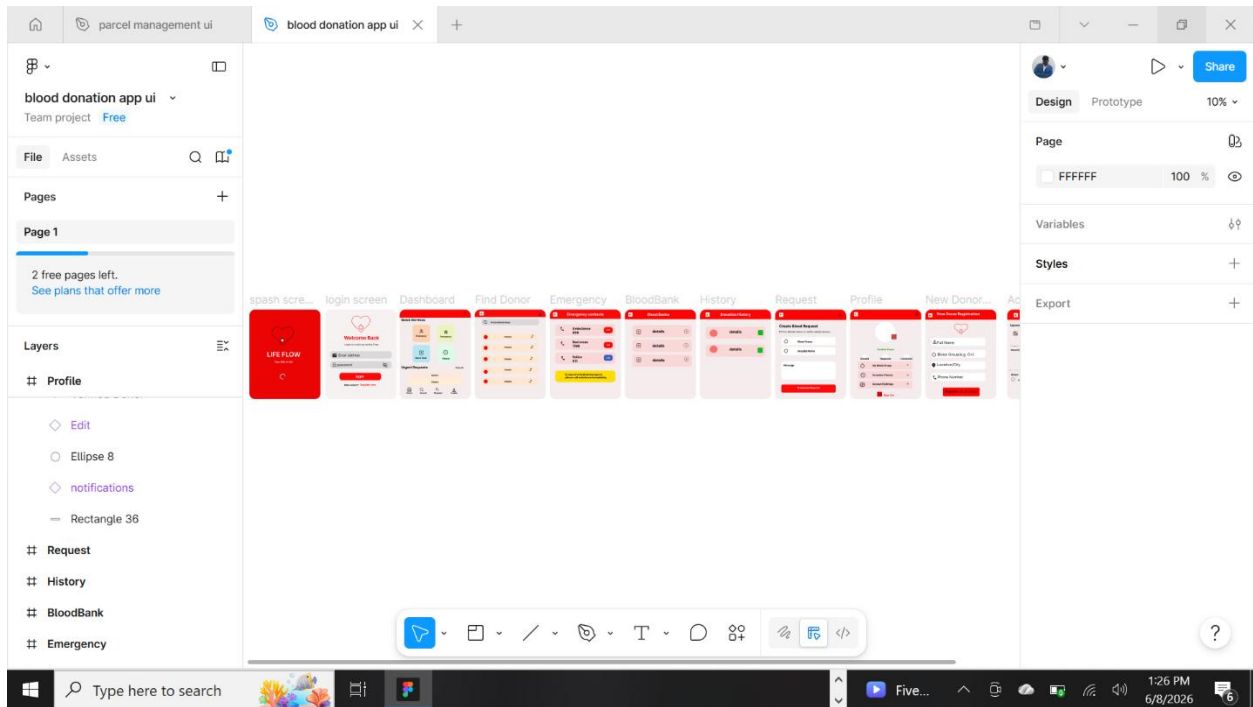
WEEK 2: MOBILE DEVELOPMENT ENVIRONMENT



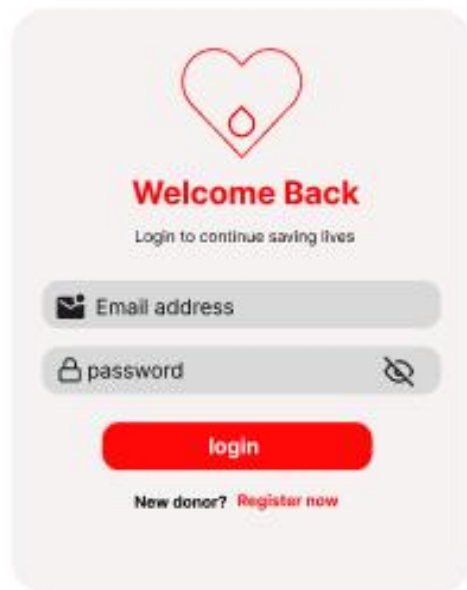
PERSONAL REFLECTION

I set up the mobile development environment by installing and configuring tools like Android Studio, Flutter SDK, and device emulators. This process helped me understand how different components work together, though I faced minor setup challenges that enhanced my troubleshooting and technical skills.

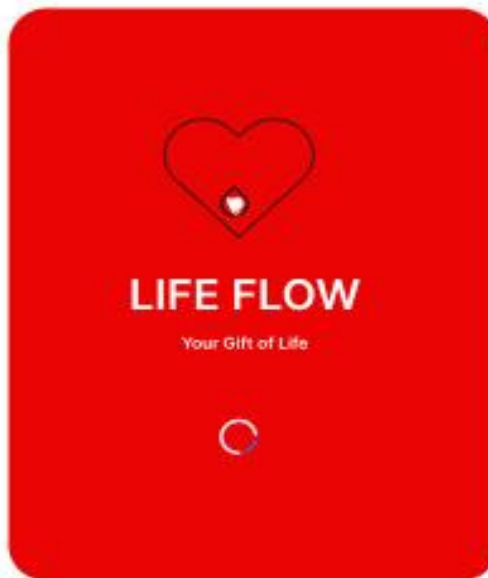
WEEK 3: USER INTERFACE DESIGN



login screen



spash screen



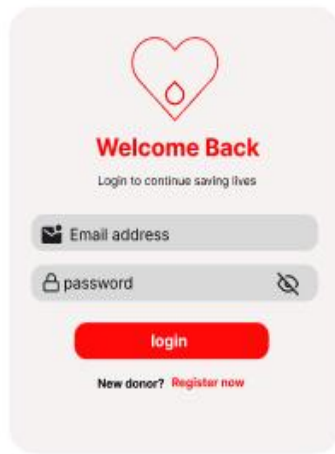


Personal reflection

I focused on user interface design for the blood donation app using Figma. I learned how to create and organize screens, design user-friendly layouts, and improve overall system usability. This improved my understanding of UI/UX principles and enhanced my design skills.

Week 4: Event handling and user interaction

login screen



The login screen features a light gray background. At the top center is a red heart icon with a drop of blood inside. Below it, the text "Welcome Back" is displayed in bold red, followed by the subtitle "Login to continue saving lives" in a smaller gray font. There are two input fields: "Email address" with an envelope icon and "password" with a lock icon and a toggle visibility icon. A red "login" button is positioned below the fields. At the bottom, the text "New donor? Register now" is shown, with "Register now" in red.

Welcome Back

Login to continue saving lives

Email address

password

login

New donor? Register now

Request



The request screen has a red header bar with a back arrow and a bell icon. The title "Create Blood Request" is in bold black, with the instruction "Fill the details below to notify nearby donors" in gray below it. There are two input fields: "Blood Group" with a drop icon and "Hospital Name" with a location pin icon. A larger "Message" input field follows. A red "Broadcast Request" button is at the bottom.

Create Blood Request

Fill the details below to notify nearby donors

Blood Group

Hospital Name

Message

Broadcast Request

18:54



Account Settings

Appearance



Dark Mode



Change Password

Old Password

New Password

Cancel

Save

Security



About



1.0.0

1 2 3 4 5 6 7 8 9 0

q w e r t y u i o p

a s d f g h j k l

⬆ z x c v b n m ⬆

!#1 , English (UK) . Done



Personal reflection

I learned about event handling and user interaction in the blood donation app. I implemented how users interact with the system through buttons, forms, and input fields. This improved my understanding of how applications respond to user actions and enhanced my programming skills.

Week 5: Data management

19:08



New Donor Registration



Full Name



Kelly maroa

Blood Group (e.g. O+)



O+

Location/City



Machakos

Phone Number



0785324646

Register as Donor



Search Blood Group



Anna Itotiah W.
Nairobi



Sharon Kendi.
Nakuru



Faith Mueni.
Mombasa



Roy Gichinga.
Kisumu



Michelle Njogu.
Eldoret



Kelley Maroa
Migori



Nicole Chepkemboi
Nandi



Home



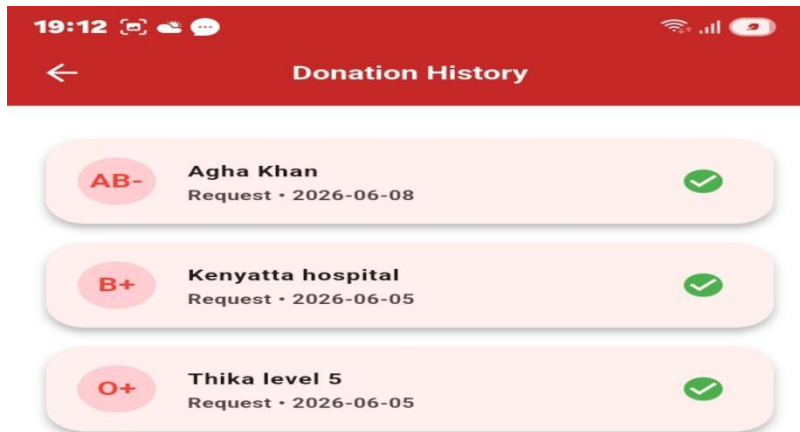
Search



Request



Profile



Personal reflection

I focused on data management in the Blood Donation App. I learned how to store, retrieve, and update donor and request information using local storage and databases. This improved my understanding of how data is handled efficiently and enhanced my skills in managing application data.

Week 6: Networking and APIs

19:17  



Account Settings

Appearance



Dark Mode

Enable dark theme for the app



Security



Change Password

Update your login credentials



About



Version

1.0.0

Password updated!

19:18



Account Settings

Appearance



Dark Mode

Enable dark theme for the app



Security



Change Password

Update your login credentials



About



1.0.0

Change Password

Old Password

New Password

Cancel

Save

19:18



Hello, Maroakelly2



Create Blood Request

Fill the details below to notify nearby donors.

Blood Group



B+

Hospital Name



Moi avenue

Message

urgent

Broadcast Request



Home



Search



Request



Profile

19:18



Hello, Maroakelly2



Create Blood Request

Fill the details below to notify nearby donors.



Blood Group



Hospital Name

Message

urgent



Broadcast Request

Request broadcasted and saved!



Home



Search

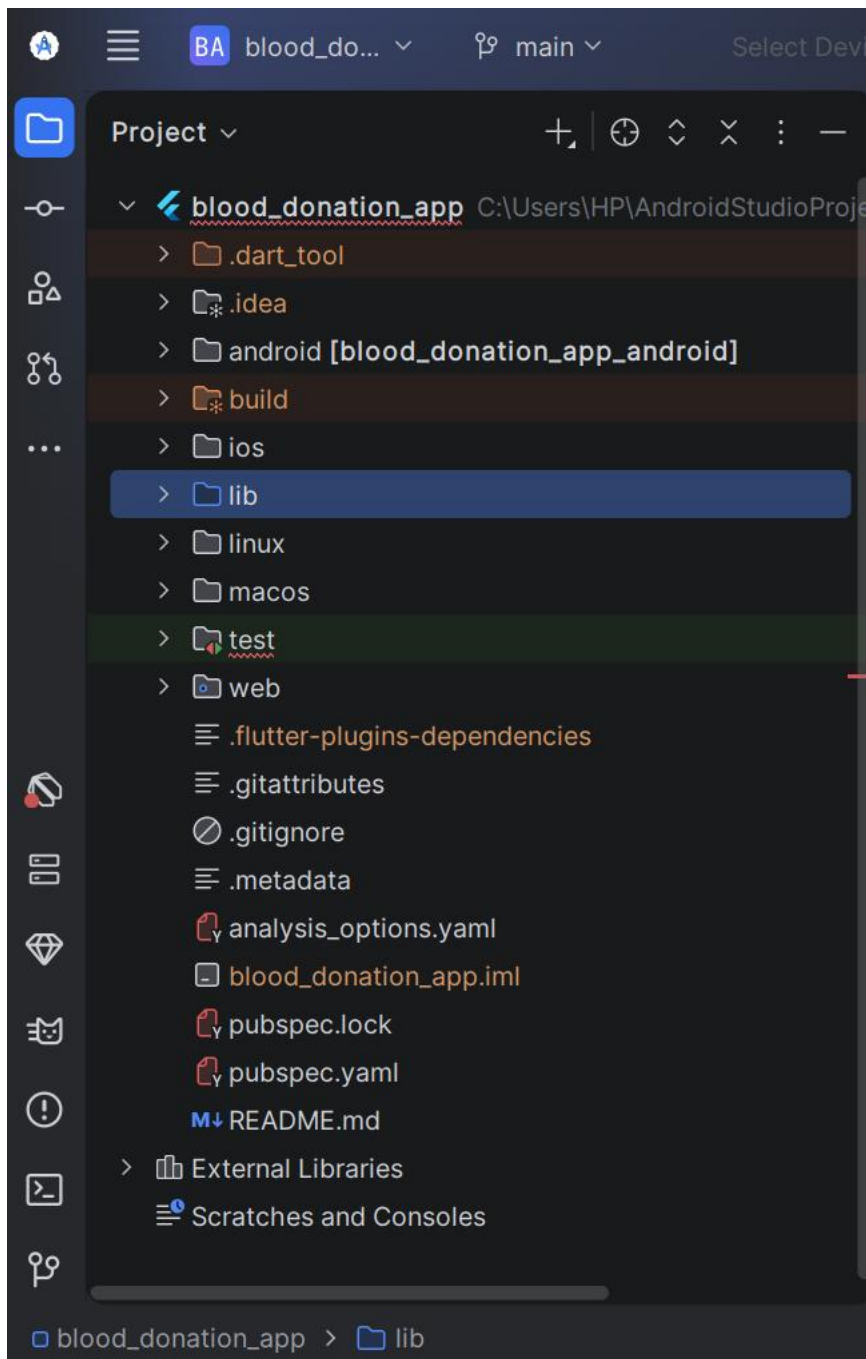


Request



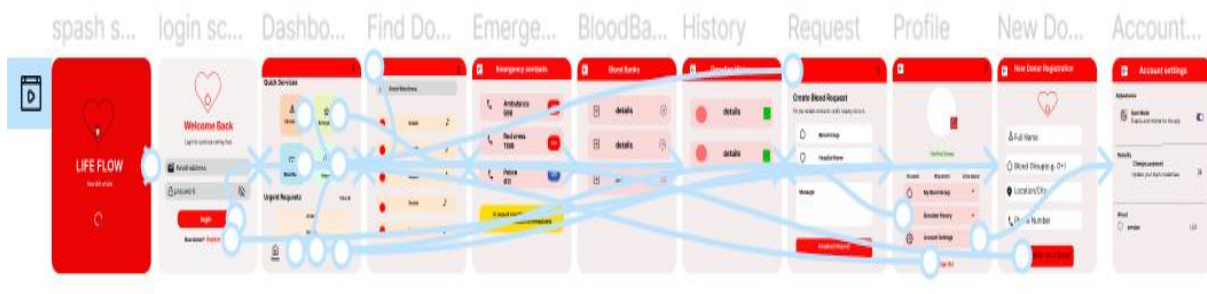
Profile

Week 7: Application Architecture



The screenshot shows the Android Studio interface with the 'blood_donation_app' project open. The left sidebar displays the project structure, including folders like 'dart_tool', 'idea', 'android', 'build', 'ios', 'lib', 'linux', 'macos', 'test', and 'web'. The main editor window shows the 'main.dart' file with the following code:

```
1 import 'package:flutter/material.dart';
2 import 'dart:async';
3 import 'dart:convert';
4 import 'package:shared_preferences/shared_preferences.dart';
5
6 void main() {
7   runApp(const ThemeWrapper());
8 }
9
10 class ThemeWrapper extends StatefulWidget {
11   const ThemeWrapper({super.key});
12
13   @override
14   State<ThemeWrapper> createState() => _ThemeWrapperState();
15
16   static _ThemeWrapperState of(BuildContext context) =>
17     context.findAncestorStateOfType<_ThemeWrapperState>();
18 }
19
20 class _ThemeWrapperState extends State<ThemeWrapper> {
21   ThemeMode _themeMode = ThemeMode.Light;
22
23   @override
24   void initState() {
25     super.initState();
```



Personal reflection

I focused on application architecture for the Blood Donation App. I learned how to structure the application into different layers such as UI, logic, and data management. This improved my understanding of how components interact within the system and helped me build a more organized and maintainable application.

WEEK 8:Project planning

Task	week
Problem identification and analysis	1-2
System design	3-4
Development/coding	5-8
Testing and Debugging	9-10
Implementation	11-12
Documentation	13
Presentation	14



Personal reflection

I focused on project planning for the Blood Donation App. I learned how to organize tasks, allocate time, and set milestones using tools like a Gantt chart. This improved my ability to manage development activities effectively and ensured the project followed a clear and structured timeline.

Week 9:Devices Features and Sensors

Emergency



Appearance



Dark Mode

Enable dark theme for the app



Security



Change Password

Update your login credentials



Personal reflection

I focused on device features and sensors in the Blood Donation App. I learned how mobile devices use features such as Darkmode

Week 9:Notifications and Background services

18:54



Account Settings

Appearance



Dark Mode

Security



About



1.0.0

Change Password

Old Password

New Password

Cancel

Save

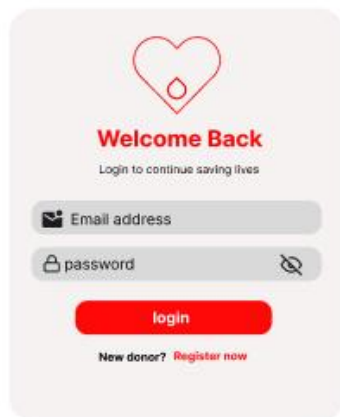


Personal reflection

I focused on notifications and background services in the Blood Donation App. I learned how the system sends alerts to users even when the app is not actively in use. This improved my understanding of real-time communication and background processing, enhancing user engagement and responsiveness of the application.

Week 11: Mobile Security

login screen



19:18



Account Settings

Appearance



Dark Mode

Enable dark theme for the app



Security



Change Password

Update your login credentials



About



Change Password

Old Password

New Password

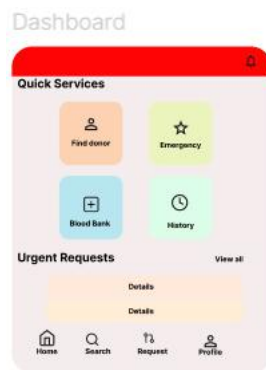
Cancel

Save

Personal reflection

I focused on mobile security in the Blood Donation App. I learned how to protect user data through authentication, secure storage, and access control mechanisms. This improved my understanding of application security and ensured that sensitive information is handled safely within the system.

Week 12: Testing and Debugging



Personal reflection

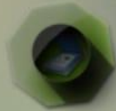
I focused on testing and debugging the Blood Donation App. I tested different system functions to ensure they work as expected and identified errors within the application. I also fixed bugs and improved system performance, which enhanced the reliability and stability of the application.

Week 13:Application Deployment

20:30



Lite



ClearScanner



ChatGPT



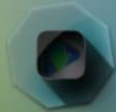
PDF Reader - Hi Read



Pinterest



WPS Office



MovieBox



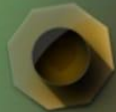
X



Reminder



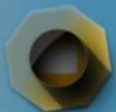
Safaricom



Locket



MyLOFT



My Files



Gallery



Photos



Snapchat



Parcel Management S...



Client management...



my_first_app



parcel_app



blood_donation_app



Vercel

Search



20:31



LIFE FLOW

Your Gift of Life



Close all



20:36



 maroakelly

blood_donation_app

☆ 0 stars  0 forks

☆ STAR



Issues

0



Pull Requests

0



Actions



Contributors

1



Watchers

0



Current branch
main



Code



Commits



README.md

EDIT



Home



Inbox



Explore



Copilot

Personal reflection

I built and tested the release APK of my Blood Donation App, deployed it on a device, and uploaded it to GitHub, gaining skills in deployment, testing, and version control.